Data Cleaning and Preprocessing of Airbnb NYC Dataset

Oasis Infobyte Internship

Domain: Data Analytics
Task: Task 4 Data Cleaning
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1. Introduction

Data cleaning is an essential step in the data analysis process. Real-world datasets often contain missing values, duplicate records, incorrect data types, and inconsistencies that can affect the accuracy of analysis and machine learning models.

The purpose of this project is to perform **data cleaning and preprocessing** on the Airbnb NYC dataset using Python. By cleaning the dataset, we ensure that the data becomes more reliable and suitable for further analysis.

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2. Objective

The main objective of this project is to clean and preprocess a dataset by identifying and handling missing values, removing duplicate records, correcting data types, and preparing the dataset for analysis.

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3. Dataset Description

The dataset used in this project is the **Airbnb NYC 2019 dataset**, which contains information about Airbnb listings in New York City.

Dataset Features

The dataset includes the following information:

- **id** – Unique identifier for each listing
- **name** – Name of the listing
- **host_id** – Unique identifier of the host
- **host_name** – Name of the host
- **neighbourhood_group** – Area of the listing
- **neighbourhood** – Specific neighborhood location
- **latitude & longitude** – Geographic coordinates
- **room_type** – Type of room offered
- **price** – Price per night
- **minimum_nights** – Minimum number of nights required
- **number_of_reviews** – Total reviews received
- **last_review** – Date of the last review
- **reviews_per_month** – Average number of reviews per month
- **availability_365** – Availability of listing throughout the year

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4. Tools and Technologies Used

The following tools and libraries were used for this project:

- **Python**
- **Google Colab**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**

These tools helped in performing data exploration, cleaning, and visualization.

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5. Methodology

The data cleaning process was performed in several steps.

Data Loading

The dataset was loaded into the Python environment using the Pandas library.

Data Exploration

Initial exploration of the dataset was performed using functions such as:

- `head()`
- `info()`
- `describe()`

These functions help understand the dataset structure and data types.

Missing Value Detection

Missing values in the dataset were identified using the `isnull()` function.

Handling Missing Values

Missing values in columns such as **name**, **host_name**, and **reviews_per_month** were handled by filling them with appropriate values.

Duplicate Removal

Duplicate rows were identified and removed to ensure the dataset contains only unique records.

Data Type Correction

Columns containing date information were converted into proper datetime format.

Dataset Export

After cleaning the dataset, the cleaned data was exported and saved as a new CSV file for further analysis.

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6. Results

After performing data cleaning operations:

- Missing values were successfully handled.
- Duplicate records were removed.
- Data types were corrected.

- The dataset became more structured and reliable.

The cleaned dataset is now suitable for further data analysis and machine learning applications.

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7. Conclusion

Data cleaning plays a critical role in the data analysis process. Without proper cleaning, datasets may contain errors that lead to incorrect results and poor model performance.

In this project, the Airbnb NYC dataset was successfully cleaned and prepared for analysis. The cleaned dataset ensures higher accuracy and reliability for future analytical tasks.

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8. Future Scope

Future improvements can include:

- Performing Exploratory Data Analysis (EDA)
- Building predictive models using the cleaned dataset
- Creating interactive visualizations and dashboards

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9. References

- Python Documentation – https://www.python.org
- Pandas Documentation – https://pandas.pydata.org
- Airbnb Dataset – Kaggle